

Assignment - 1

Course: MST501-1: Python
Programming for Statistics



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Q1. How to Download and Install Python?

Ans- The steps to download and install Python are as follows-

Step 1- We open a web browser and visit the official Python website, <https://www.python.org> and click on the Downloads option.

Step 2: We open the downloaded Python installer and select the option "Add Python to PATH" then click Install Now. We wait for the installation process to complete and click on close after the installation is successful.

Step 3: We press Windows +R and type "cmd" and press enter to open the command prompt. We then type

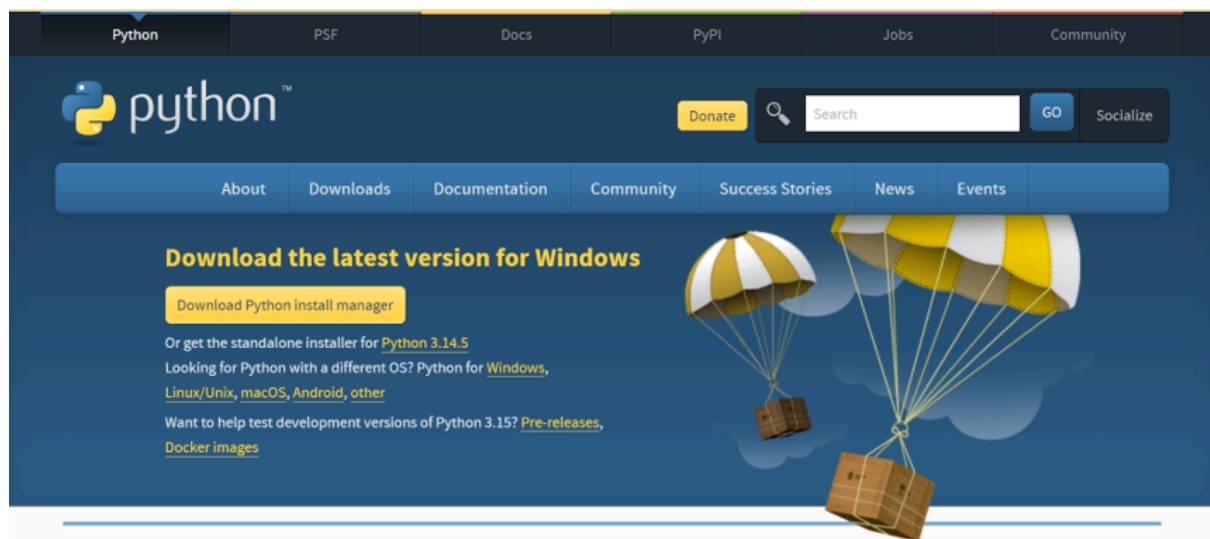
```
python --version
```

And press Enter.

After the Python is installed correctly, the version number will be displayed.

Example:

Python 3.13.0



Q2. How a Python Program is Executed? Explain the Steps from Creation to Run with Diagram and Example.

Ans- Python is both a compiled and interpreted language and first the source code is compiled into bytecode, and then the bytecode is interpreted by the Python Virtual Machine (PVM).

The steps are-

Step 1-The programmer writes the Python code and saves it with the .py extension.

Example:

```
a = 10
```

```
b = 20
```

```
c = a + b
```

```
print("Sum =", c)
```

The program is saved as:

filename.py

Step 2- When the program is executed, the Python compiler checks the code for syntax errors and converts it into bytecode and the bytecode file is saved with the .pyc extension.

Example:

filename.py → filename.pyc

And the compiler takes the Python source code as input and generates bytecode as output.

Step 3- The generated bytecode cannot be executed directly by the computer. Therefore, it is passed to the Python Virtual Machine (PVM) which then interprets the bytecode and converts it into machine code that the computer's processor can understand and execute.

Step 4- After the machine code is executed, the result is displayed on the screen.

Example:

```
a = 10
```

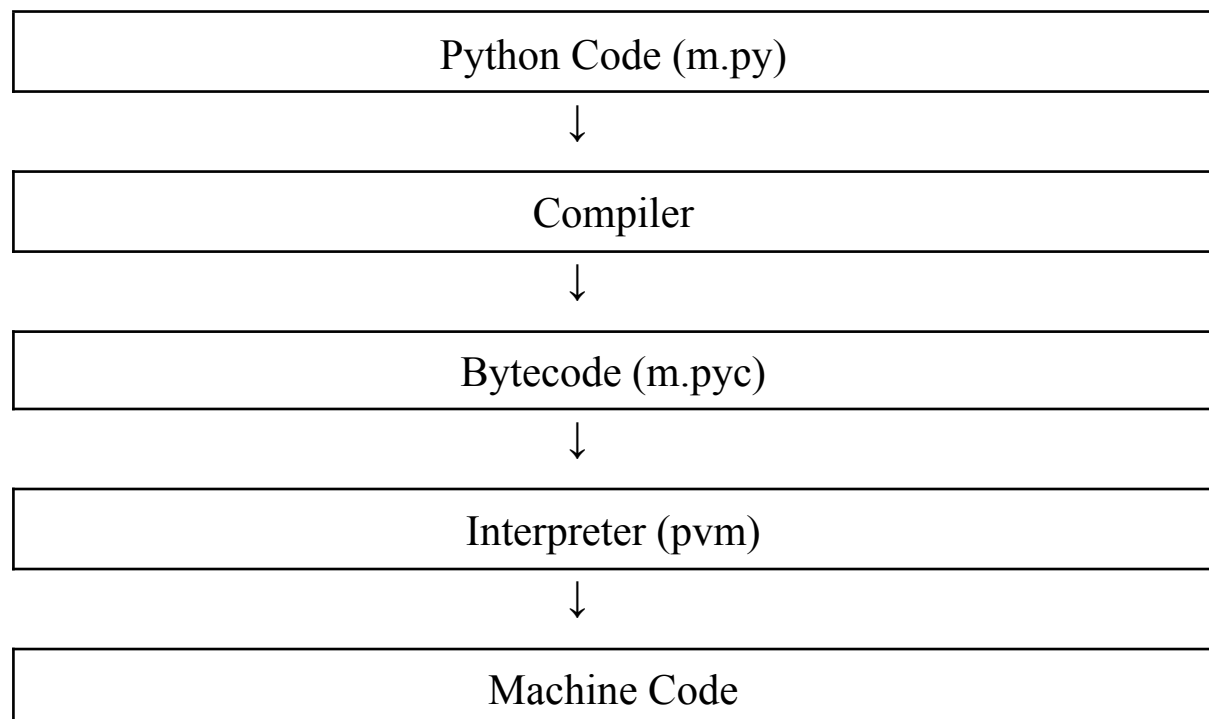
```
b = 20
```

```
c = a + b
```

```
print("Sum =", c)
```

Output:

```
Sum = 30
```



Q3. Write a python code for "Personalized SMART Water Bottle". Take user input as their Name and age. Set a target for water consumption according to age. Take inputs from user about their water intake. Print personalized messages according to their progress. e.g. if their target achieved congratulations, if near, the motivation and also mention the amount left, and so on.

```
Ans- name=input("Enter your Name")
```

```
age=int(input("What is your age"))
```

```
if age<10:
```

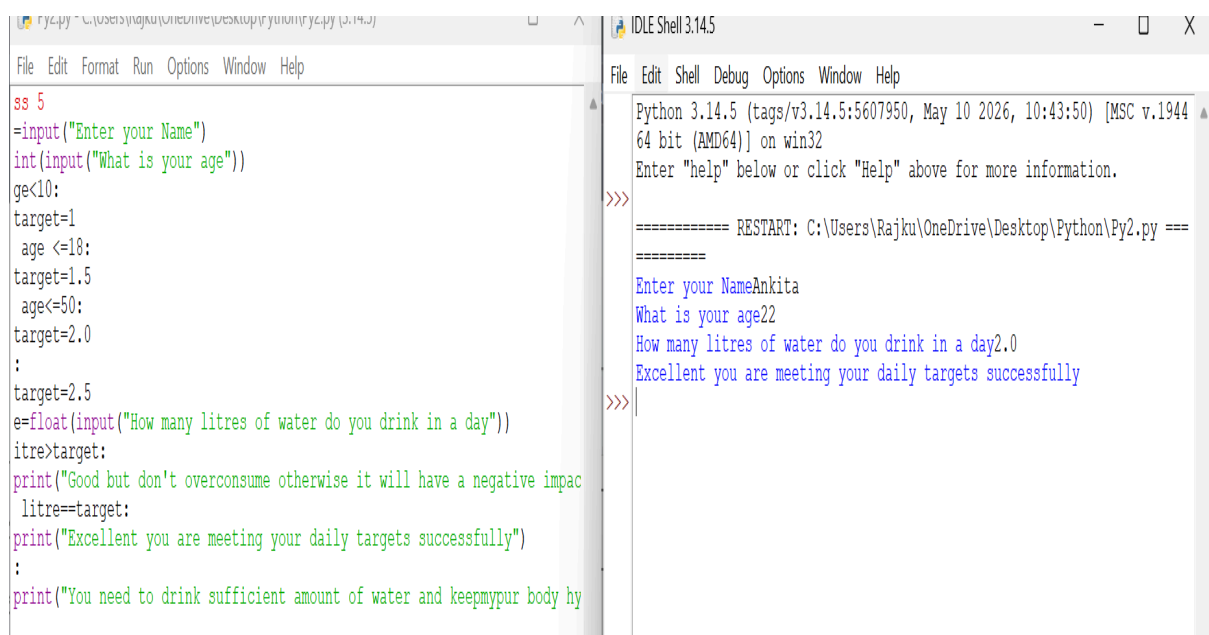
```
    target=1
```

```
elif age <=18:
```

```

    target=1.5
elif age<=50:
    target=2.0
else:
    target=2.5
litre=float(input("How many litres of water do you drink in a day"))
if litre>target:
    print("Good but don't overconsume otherwise it will have a negative impact
on your health")
elif litre==target:
    print("Excellent you are meeting your daily targets successfully")
else:
    print("You need to drink sufficient amount of water and keepmypur body
hydrated")

```



The screenshot shows a Python IDE with two windows. The left window displays the source code for a program that asks for a user's name, age, and daily water intake, then checks if they meet certain targets. The right window shows the execution output, where the user has entered 'Ankita' for their name, '22' for their age, and '2.0' for their daily water intake. The program has successfully executed and printed the message 'Excellent you are meeting your daily targets successfully'.

```

ss 5
= input("Enter your Name")
int(input("What is your age"))
ge<10:
target=1
age <=18:
target=1.5
age<=50:
target=2.0
:
target=2.5
e=float(input("How many litres of water do you drink in a day"))
itre>target:
print("Good but don't overconsume otherwise it will have a negative impac
litre==target:
print("Excellent you are meeting your daily targets successfully")
:
print("You need to drink sufficient amount of water and keepmypur body hy

```

```

Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944
64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py ===
=====
Enter your NameAnkita
What is your age22
How many litres of water do you drink in a day2.0
Excellent you are meeting your daily targets successfully
>>>

```

Output-

Enter your NameAnkita

What is your age22

How many litres of water do you drink in a day2.0

Excellent you are meeting your daily targets successfully

